

$$2x^2y^2z^2 + 2x^2yz^2 + 2xy^2z^2 + 2xyz^2 + 2x^2yz + 2xy^2z + 2xyz = \Sigma(0, 2, 4, 6, 7)$$

Handwritten Karnaugh map for the function $f(A, B, C, D) = \sum(0, 1, 2, 3, 4, 5, 6, 7)$. The map is a 4x4 grid with columns labeled 00, 01, 11, 10 and rows labeled 2, 0, 1. The cells contain 1s for minterms 0, 1, 2, 3, 4, 5, 6, and 7. The bottom-right cell (row 1, column 10) is empty.

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$$F' = x'z + y'z$$

$$F = (F')' = (x'z + y'z)'$$

$$\begin{aligned} \mathbb{F}' &= (x'z + y'z) \\ &= (x'z)' + (y'z)' \quad (\text{Theorem 5(a)}) \\ &= (x'z)' + (y'z)' \quad (\text{Theorem 5(b)}) \end{aligned}$$

$$= (x+z) \cdot (y+z) \quad (\text{Theorem 3}) \rightarrow \text{Simplified POS}$$

$$\begin{aligned}
 &= (A+B'D) \cdot (A+B'D) \cdot (C'D) \cdot (C'D') \\
 (b) & (A+B'D) \cdot (A+B'D) \cdot (C'D) \cdot (C'D') \quad (\text{Postulate } 2(a)) \\
 &= (A+B'D) \cdot (A+B'D) \cdot ((C'D)+0) \cdot ((C'D')+0) \quad (\text{Postulate } 5(b)) \\
 &= (A+B'D) \cdot (A+B'D) \cdot ((C'D)+A \cdot A') \cdot ((C'D')+A \cdot A') \quad (\text{Postulate } 4(b)) \\
 &= (A+B'D) \cdot (A+B'D) \cdot (C'D+A) \cdot (C'D'+A) \cdot (C'D+A') \cdot (C'D'+A') \quad (\text{Postulate } 4(b)) \\
 &= (A+B'D) \cdot (A+B'D) \cdot (C'D+A) \cdot (C'D'+A) \cdot (C'D+A+0) \cdot (C'D'+A+0) \\
 &= (A+B'D+0) \cdot (A+B'D+0) \cdot (C'D+A+0) \cdot (C'D'+A+0) \cdot (C'D+A'+B \cdot B') \\
 &\quad \cdot (C'D'+A'+B \cdot B') \quad (\text{Postulate } 2(a)) \\
 &= (A+B'D+C \cdot C') \cdot (A+B'D+C \cdot C') \cdot (C'D+A+B \cdot B') \cdot (C'D'+A'+B \cdot B') \\
 &\quad \cdot (C'D+A+B) \cdot (C'D'+A'+B) \quad (\text{Postulate } 5(b)) \\
 &= (A+B'D+C) \cdot (A+B'D+C) \cdot (A+B'D+C) \cdot (A+B'D+C) \cdot (C'D+A+B) \cdot (C'D'+A'+B) \\
 &= (A+B'D+C) \cdot (A+B'D+C) \cdot (A+B'D+C) \cdot (A+B'D+C) \cdot (C'D+A+B) \cdot (C'D'+A'+B) \\
 &\quad \cdot (C'D+A+B) \cdot (C'D'+A'+B) \quad (\text{Postulate } 4(b)) \\
 &= (C'D+A+B) \cdot (C'D'+A'+B) \quad (\text{Postulate } 4(b))
 \end{aligned}$$